

Innovation survey metadata

Wave 2016-2018

This document provides quantitative and qualitative information on the design and implementation of national innovation surveys covering details such as survey sectoral coverage, weights calculation method, response rate, and imputation methods. It refers to the 2021 OECD data collection of innovation indicators with reference period 2016-2018 or closest. As on previous occasions, such information is collected to build a comprehensive picture of the key changes that have been introduced in surveys over time and help us draw more precise lessons about the impact of survey design on data quality and international comparability.

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Australia

Name of survey	Business Characteristics Survey (BCS) 2018/19. The sectoral and size coverage of enterprises matches the CIS scope.
Name of responsible agency	Australian Bureau of Statistics
Reference period	2018-2019, (fiscal years)
Length of reference period	1 year (The BCS is an annual survey, however only every second cycle the additional detailed innovation module is included).
Reference year	2018-2019
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions.
Differences with the OSLO manual	No differences to Oslo Manual concepts and definitions.
Difference with the CIS questionnaire	The CIS model questionnaire is used as a starting point for innovation content on the BCS. Content is also determined by user demand and form testing to see which questions can be answered accurately.
Target population cut-off point	All employing businesses are in the sample.
Industry coverage	The scope of the estimates in this publication consists of all employing business entities in the Australian economy, except for: SISCAS 3000 General government SISCAS 6000 Rest of the world ANZSIC06 Division A Agriculture, forestry and fishing ANZSIC06 Division O Public administration and safety ANZSIC06 Division P Education and training ANZSIC06 Groups 624 (Financial asset investing) and 633 (Superannuation funds) ANZSIC06 Groups 854 (Religious services) and 955 (Civic, professional and other interest group services) ANZSIC06 Subdivision 96 Private households employing staff
Sampling frame used	National statistical business register
Completion requirement	Compulsory, with enforceable penalties
Unit of analysis surveyed	Businesses with a simple structure: Most businesses and organisations in Australia need to obtain an Australian Business Number (ABN). They are then included on the whole-of-government register of businesses, the Australian Business Register (ABR), which is maintained by the Australian Taxation Office (ATO). Most of these businesses have simple structures; therefore, the unit registered for an ABN will satisfy ABS statistical requirements. For these businesses, the ABS has aligned its statistical units structure with the ABN unit. The businesses with simple structures constitute the ATO maintained population (ATOMP), and the ABN unit is used as the statistical unit for all ABS economic collections. Businesses with a complex structure: For the population of businesses where the ABN unit is not suitable for ABS statistical requirements, the ABS maintains its own units structure through direct contact with the business. These businesses constitute the ABS maintained population (ABSMP). This population consists typically of large, complex and diverse businesses. For businesses in the ABSMP, statistical units comprise the Enterprise Group, the Enterprise and the Type of Activity Unit (TAU). The range of activities across the Enterprise Group can be very diverse. The TAU represents a grouping of one or more business entities within the Enterprise that cover all of the operations within an industry subdivision and for which a basic set of financial production and employment data can be reported.
Survey type	Sample survey
Type of interviews	Webform and Postal questionnaire
Average time to complete the questionnaire	1 hour 20 min
Target population	6979

Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	90%
Definition of strata	The sample was stratified by industry and an employment-based size indicator.
Weights calculation methods: variables used and methodology	For the BCS, the weights are calculated with respect to the following factors: the probability of selection for each survey unit (probability weighting); adjustments to account for problems with the survey frame - such as missing units (new business provisions); and adjustment for non-response - to correct for further imbalances in the characteristics of responding sample units (post-stratification).
Implementation of non-response survey	
Number and types of reminders	4 (mailed out letters)
Use of imputation methods	Categorical items that are missing are not imputed. An adjustment for missing categorical variables is made as part of calculating the proportions. A check is made of response rates to each question and where the response rate is low enough to impact on output quality, some follow-up of missing responses is undertaken. There are some items (such as the key innovation and IT indicators) where a response must be obtained. Fully non-responding units are implicitly imputed using weight adjustment, i.e. the weight of other units in the stratum is adjusted upwards to account for the non-respondents. Few partial responses (13 units) were imputed using Live Respondent Mean (LRM) only for selected variables.
Combined with other surveys	Yes (IT and Business Characteristics Survey)
Link to documentation (outputs and questionnaire)	Characteristics of Australian Business, 2018-19 financial year Australian Bureau of Statistics (abs.gov.au)
Survey scope includes public enterprises (i.e. central, regional or local government-controlled) (Yes/no/explain)	

Austria

Name of survey	Community Innovation Survey (CIS 2018)
Name of responsible agency	Survey is funded/commissioned by the Federal Ministry of Digital and Economic Affairs (BMDW); all work (concept, data collection, analyses) is done by the NSO (Statistics Austria).
Reference period	2016-2018
Reference year	2018
Length of reference period	3 years
Data collection period	October 2019 - April 2020
First release of innovation results	end of June 2020
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions, like on Eurostat's CIS model questionnaire.
Differences with the OSLO manual	Questionnaire design is extremely close to the Eurostat core questionnaire which is in line with the Oslo Manual definitions.
Difference with the CIS questionnaire	None. Not all questions of the CIS questionnaire were asked.
Target population cut-off point	10 or more employed persons
Industry coverage	ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Voluntary
Unit of analysis surveyed	Enterprise (= legal unit)
Survey type	Stratified random sample. Enterprises with 250 and more employed persons were subject to a census; small and medium enterprises were selected via a stratified random sample.
Type of interviews	Web and postal questionnaire
Average time to complete the questionnaire	Unknown. This information is not collected for voluntary surveys.
Target population	18,101
Target population (manufacturing)	6,664
Target population (services)	10,852
Achieved sample	2,793
Achieved sample (manufacturing)	1,133
Achieved sample (services)	1,551
Weighted unit response rate	n.a
Unweighted unit response rate	51.8
Definition of strata	For the sampled enterprises a stratified random sampling was applied. The variables used for the stratification of the sample were the economic activity of the enterprise according to NACE Rev.2 classification, enterprise size and NUTS 1-region.
Weights calculation methods: variables used and methodology	Iterative proportional fitting was used to adjust the weights according to the number of enterprises in the strata and the number of innovative enterprises by size class.
Implementation of non-response survey	A non-response survey was carried out among 50% of all non-responding firms (=1,519 enterprises). A random sample, stratified by size - 3 classes - and NACE - 3 classes (manufacturing, wholesale, other services) - was drawn.
Number and types of reminders	In total three letters were sent to enterprises. One letter informing about the launch of the survey and two additional reminders. Both reminders included a paper questionnaire to encourage survey participation.

Use of imputation methods	Hotdeck methods and use of information from previous R&D surveys for R&D information.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	http://www.statistik.at/web_de/statistiken/energie_umwelt_innovation_mobilitaet/forschung_und_innovation/innovation_im_unternehmenssektor/index.html http://www.statistik.at/web_de/dokumentationen/energie_umwelt_innovation_mobilitaet/ForschungFEInnovation/index.html
Survey scope includes public enterprises (i.e. central, regional or local government-controlled) (Yes/no/explain)	Yes

Belgium

Name of survey	Community Innovation Survey 2018
Name of responsible agency	Belgian Science Policy Office (https://meri.belspo.be/site/index_en.stm)
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	From approx. April 2019 until October 2019
First release of innovation results	June 30, 2020
Concepts, notes and definitions in the questionnaire	Diagrams and notes explaining innovation concepts immediately precede or follow the questions they relate to. Links for websites with general information on Nace-codes or examples of innovations are provided within the different options response options.
Differences with the OSLO manual	None
Difference with the CIS questionnaire	We covered all the compulsory core Nace sectors. However, even within the compulsory sectors, some cells are missing. This is because no observations were available for these cells (either because there is no firm in the population, or because none of the surveyed firms answered the questionnaire). Belgium is a small country, therefore we also run into confidentiality issues for certain cells. We did not ask the following voluntary questions of the Eurostat model questionnaire: the questions on outlook (5.3), information sources (7.1), the effect of legislation on innovation (11.1 and 11.2), the reasons for no innovation (12.1, 12.2, 12.3), intellectual property rights (13.1), nor the module on logistics (section 14).
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev. 2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73
Sampling frame used	Due to confidentiality constraints the official Belgian business register could not be used. Instead, we used the register available from the Belgian National Social Security Office (which contains all active employers in Belgium) as frame population. This official register is at the level of legal units. This register was agreed upon by Statistics Belgium as being statistically equivalent to the official business register.
Completion requirement	Voluntary
Unit of analysis surveyed	Legal units with an enterprise number (i.e. VAT numbers) in the national business register
Survey type	Combination of sample survey and census of the enterprises included in the frame population. The employment size class is used to define a threshold. Usually medium-size (50-249 employees) and large (250 or more employees) enterprises are completely enumerated, while smaller enterprises are randomly sampled using a stratified design. More details are available in the Quality Report submitted to Eurostat.
Type of interviews	Web, postal questionnaire, and phone interviews
Average time to complete the questionnaire	We don't ask for this in the questionnaire
Target population	14515
Target population (manufacturing)	5407
Target population (services)	8755
Achieved sample	4984
Achieved sample (manufacturing)	2036
Achieved sample (services)	2819
Weighted unit response rate	62%
Unweighted unit response rate	61%

Definition of strata	For the randomly sampled enterprises a stratified random sampling design was applied. The variables used for the stratification of the sample were the economic activity of the enterprise (according to ISIC Rev. 4/NACE Rev.2 classification) and the enterprise size (10-49, 50-249, 250+). In one region the R&D status was also used in their stratification design. More details are available in the Quality Report submitted to Eurostat.
Weights calculation methods: Variable used and methodology	Weights are simply N/n , the number of firms in the population over the number of firms in the achieved sample
Implementation of non-response survey	Only partially (one region out of the three), but results are not used, considering a randomized experiment demonstrated that consistently higher innovation rates are obtained when abbreviated forms are used to survey innovation, the practice of conducting non-response surveys with abbreviated forms was abandoned in most regions.
Number and types of reminders	Two reminders were sent by postal mail. Follow-up phone calls were started after the first reminder.
Use of imputation methods	We used the SAS application provided by Eurostat in the context of CIS 4. We updated it to fit the current version of CIS. Imputation of metric variables (turnover, innovation expenditure, share of turnover due to products new to the market/to the firm only etc.) was done using weighted ratio means. Imputation of nominal and ordinal variables was done using nearest neighbor hot-decking.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	National figures and a short analysis can be found here: https://meri.belspo.be/site/innovation_en.stm Regional results: https://www.ecoom.be/nodes/cisrapport/en ; https://www.vlaamsindicatorenboek.be/4.4/innovatie-inspanningen-van-ondernemingen ; https://recherche.wallonie.be/go/INNO-fr Survey forms are available upon request, contact us at: http://meri.belspo.be/stie/contact_en.stm .
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	No. Only market oriented public enterprises are included, which is in line with the Oslo Manual and Frascati Manual guidelines. The number of such market oriented public enterprises is fairly small. We do not measure public sector innovation in general.

Brazil

Name of survey	2017 Brazilian Innovation Survey
Name of responsible agency	Brazilian Institute of Geography and Statistics – IBGE
Reference period	2015-2017
Length of reference period	3 years
Reference year	2017
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	
Differences with the OSLO manual	
Difference with the CIS questionnaire	
Target population cut-off point	enterprises with at least 20 employees and \$ 250,000 annual operating revenue in 2017
Industry coverage	Classification NAICS (2017): Sectors: 11, 21, 22, 23, 31, 32, 33, 41, 44, 45, 48, 49, 51, 52, 53, 54, 55, 56
Sampling frame used	The sectoral coverage does not include ISIC Rev. 4/NACE Rev.2 sections E and H. The size coverage of enterprises matches the CIS scope.
Completion requirement	
Unit of analysis surveyed	Enterprise
Survey type	
Type of interviews	
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	
Number and types of reminders	
Use of imputation methods	
Combined with other surveys	
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises” (Yes/no/explain)	

Canada

Name of survey	2019 Survey of Innovation and Business Strategy
Name of responsible agency	Statistics Canada
Reference period	2017-2019
Length of reference period	3 years
Reference year	2019
Data collection period	January – March 2020
First release of innovation results	April 26, 2021
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions
Differences with the OSLO manual	In line with Oslo Manual, 4th edition.
Difference with the CIS questionnaire	The industries surveyed are different from the CIS core coverage and include the following 2017 NAICS divisions: 11, 21, 22, 23, 31, 32, 33, 41, 44, 45, 48, 49, 51, 52, 53, 54, 55 and 56.
Target population cut-off point	Minimum of 20 employees with revenues of at least \$250,000
Industry coverage	The industries surveyed include the following 2017 NAICS divisions: 11, 21, 22, 23, 31, 32, 33, 41, 44, 45, 48, 49, 51, 52, 53, 54, 55 and 56.
Sampling frame used	The Business Register (BR) was used as the frame. The BR is a data service centre updated through a number of sources including administrative data files, feedback received from conducting Statistics Canada business surveys and profiling activities including direct contact with companies to obtain information about their operations and Internet research findings.
Completion requirement	Mandatory
Unit of analysis surveyed	Enterprise
Survey type	Combined (Census of large enterprises, those with at least 250 employees; and-Sample of small and medium enterprises to expect the production of estimates with an expected standard error no greater than 10% for percent estimates at the national level).
Type of interviews	Electronic questionnaire (In some cases, respondents completed the questionnaire over the phone with an interviewer).
Average time to complete the survey	21 minutes
Target population	Minimum of 20 employees with revenues of at least \$250,000 belonging to one of the 14 industry sectors in NAICS 2017
Target population (manufacturing)	Minimum of 20 employees with revenues of at least \$250,000 belonging the industry sectors 31-33 in NAICS 2017
Target population (services)	Minimum of 20 employees with revenues of at least \$250,000 belonging the industry sectors 41, 44-45, 48-49, 51, 52, 53, 54, 55, 56 in NAICS 2017
Achieved sample	14,985 enterprises
Achieved sample (manufacturing)	3,963 enterprises
Achieved sample (services)	9,094 enterprises
Weighted unit response rate	
Unweighted unit response rate	75.3%
Definition of strata	Random sample methodology stratified by industry, region and enterprise size (3 enterprise classes based on number of employees, 20-99, 100-249, +250 employees).
Weights calculation methods: Variables used and methodology	A complete file of weighted micro data was created for all sampled enterprises in the survey population for which data were reported or imputed. Weights were adjusted by a factor to account for total non-response so that the final estimates would be representative of the entire survey population. Weighted estimates were produced using the Generalised System of Estimation.

Implementation of non-response survey	Imputation is performed only to treat item non-responses.
Number and types of reminders	Intensive non-response follow-up was conducted by email, telephone and fax as appropriate.
Use of imputation methods	The imputation of item non-responses is performed using the nearest neighbour donor imputation procedure in the generalized system BANFF. This procedure finds, for each record requiring item imputation, its most similar valid record thereby allowing the imputed recipient record to pass the specified imputation edits and post edits rules.
Combined with other surveys	Contains modules on business strategy and global value chains.
Link to documentation (outputs and questionnaire)	https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=5171
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	Yes - provided that the enterprise is in one of the 14 NACIS 2017 industry groups, has a minimum of 20 employees with revenues of at least \$250,000.

Chile

Name of survey	Chilean Business Innovation Survey
Name of responsible agency	INE Chile, (survey design and estimation carried out in conjunction with the Ministry of Science, Technology, Knowledge and Innovation)
Reference period	2017-2018
Length of reference period	2 years
Reference year	2018
Data collection period	2019
First release of innovation results	First semester 2020
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions. The annex and footnotes provide general details and examples.
Differences with the OSLO manual	
Difference with the CIS questionnaire	Questions, sequencing, type of categories, form of questions.
Target population cut-off point	
Industry coverage	Covers enterprises above 2.400 UF (Unidades de Fomento for its name in Spanish) annual sales (equivalent to US \$ 103.326 dollars of the year 2018) . The sectoral coverage includes all ISIC Rev. 4 sections except P, Q and S.
Sampling frame used	Enterprises sample frame owned by INE Chile and created from tax and others survey data.
Completion requirement	Compulsory, with enforceable penalties
Unit of analysis surveyed	Enterprise
Survey type	Combined. For mining and quarrying, and Electricity is Census, (these enterprises are included directly in the sample). The others sectors are sampled.
Type of interviews	Self-applied by web. There's email and telephone follow-ups which include help for answering the survey when needed.
Average time to complete the survey	
Target population	All enterprises wit annual sales higher than 2.400 UF (Unidad de Fomento for its name in Spanish), equivalent to US \$ 103.326 dollars of the year 2018.
Target population (manufacturing)	Equal for all sections included
Target population (services)	Equal for all sections included
Achieved sample	5.961 units
Achieved sample (manufacturing)	1.417
Achieved sample (services)	1.834
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	Variables: sector, size and region. Methodology: Based of selection probability, corrected by strata reclassification and a smoothing process. For more details see: shorturl.at/tELNU or shorturl.at/pwMW2
Implementation of non-response survey	
Number and types of reminders	Reminders by phone and email. The optimal number of contacts was 3 and considers only the direct contact between the analyst and the informant. However, to reach the informant can take up to seven attempts.
Use of imputation methods	No imputations were made.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	https://observa.minciencia.gob.cl/encuesta/encuesta-nacional-de-innovacion

Survey scope includes public (i.e. central,
regional or local government-controlled)
enterprises" (Yes/no/explain)

China

Name of survey	Chinese Innovation Survey
Name of responsible agency	National Bureau of Statistics of China
Reference period	2018
Length of reference period	1 year
Reference year	2018
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	
Differences with the OSLO manual	In line with Oslo Manual, 3rd edition.
Difference with the CIS quest.	
Target population cut-off point	
Industry coverage	According to Industrial Classification for National Economic Activities of China (GB/T 4754—2017, the sectoral coverage includes Mining(B), Manufacturing(C), Production and Supply of Electricity, Heating, Gas and Water(D), Construction(E), Wholesale and Retail Trades(F), Transport, Storage and Post(G), Information Transmission, Software and Information Technology(I), Leasing and Business Services(L), Scientific Research and Technical Services(M), and Management of Water Conservancy, Environment and Public Facilities(N).
Sampling frame used	
Completion requirement	
Unit of analysis surveyed	Enterprise
Survey type	Census of enterprises above designated size, and sample survey of enterprises below designated size.
Type of interviews	Online survey
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	By industry, region and enterprise size.
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	
Number and types of reminders	
Use of imputation methods	
Combined with other surveys	Enterprise R&D Survey
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled enterprises" (Yes/no/explain)	

Czech Republic

Name of survey	Eurostat Community Innovation Survey 2018 and National Innovation Survey
Name of responsible agency	Czech Statistical Office
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	March 2019-October 2019
First release of innovation results	April 2019
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions.
Differences with the OSLO manual	No differences
Difference with the CIS questionnaire	Selected questions are adopted (includes all CIS compulsory variables), same sequencing
Target population cut-off point	10 or more persons employed
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Compulsory, with enforceable penalties
Unit of analysis surveyed	Enterprise
Survey type	Combined
Type of interviews	Electronic (PDF, WEB) or postal questionnaire
Average time to complete the survey	1,7 hours (estimated)
Target population	24 265
Target population (manufacturing)	12 194
Target population (services)	11 114
Achieved sample	5 749
Achieved sample (manufacturing)	3 355
Achieved sample (services)	1 937
Weighted unit response rate	88,8%
Unweighted unit response rate	89,4%
Definition of strata	By size class, NACE and NUTS2
Weights calculation methods: Variables used and methodology	Weights were calibrated to number of enterprise, number of persons employed according to business register and value of sales according to VAT declaration. Generalized linear regression (GREG) was used.
Implementation of non-response survey	No non-response survey is carried out.
Number and types of reminders	Two postal reminders are usually used. If necessary, the second reminder is followed by email and telephone follow-ups.
Use of imputation methods	Number of persons employed and turnover were covered by SBS survey and/or Labour cost survey. For units with non-response imputation was used. For turnover in first step data from Income tax statement were used, if not available regression model with sales according to VAT declaration as auxiliary variable was used. For number of persons employed as auxiliary variable a number of persons employed according to Business register was used.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled enterprises" (Yes/no/explain)	

Denmark

Name of survey	Eurostat Community Innovation Survey 2018 and National Innovation Survey
Name of responsible agency	Statistics Denmark
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions. The annex and footnotes contain general details and examples.
Differences with the OSLO manual	
Difference with the CIS questionnaire	Adopted almost in total (includes all CIS compulsory variables)
Target population cut-off point	
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Compulsory, with enforceable penalties
Unit of analysis surveyed	Enterprise
Survey type	Combined
Type of interviews	Web questionnaire
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	By size class and NACE
Weights calculation methods:	
Variables used and methodology	
Implementation of non-response survey	No non-response survey is carried out.
Number and types of reminders	
Use of imputation methods	
Combined with other surveys	Yes (R&D survey)
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	

Estonia

Name of survey	Eurostat Community Innovation Survey 2018 and National
Name of responsible agency	Statistical Office of Estonia
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	In the annex or in the footnotes and repository of definitions is online.
Differences with the OSLO manual	None
Difference with the CIS questionnaire	Practically identical
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Compulsory, with enforceable penalties
Unit of analysis surveyed	Enterprise
Survey type	
Type of interviews	Web questionnaire
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	For the sampled enterprises a stratified random sampling was applied. The variables used for the stratification of the sample were the economic activity of the enterprise (according to NACE Rev.2 classification - see question 17 for industry coverage) and the enterprise size (10-19, 20-49, 50-99, 100-249, 250+).
Weights calculation methods:	Variables: Number of enterprises
Variables used and methodology	Methodology: Weight for every stratum = number of respondents with filled questionnaire / number of firms in target population
Implementation of non-response survey	No non-response survey was carried out.
Number and types of reminders	
Use of imputation methods	Imputation of main variables: R&D expenses from R&D survey, turnover from SBS survey
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	

Finland

Name of survey	Eurostat Community Innovation Survey 2018 and National Innovation Survey (Innovation activity in enterprises 2016–2018)
Name of responsible agency	Statistics Finland
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	August 2019–January 2020
First release of innovation results	23 April 2020
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions, like on the Eurostat's CIS harmonised questionnaire.
Differences with the OSLO manual	None
Difference with the CIS questionnaire	CIS harmonised questionnaire mainly adopted, some voluntary questions dropped. Few national questions on topical issues (collaboration between enterprises and universities, skills and competence areas required and needed by enterprises, and the questions on the effects of covid19) were added.
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Compulsory, but not enforceable in practice
Unit of analysis surveyed	Enterprise
Survey type	Combined (Combination of sample survey and complete enumeration of enterprises included in the frame population. Large enterprises are enumerated, small enterprises are sampled. The threshold is 250 employees).
Type of interviews	Web questionnaire (possibility to print paper questionnaire out from Statistics Finland web page)
Average time to complete the survey	-
Target population	8840
Target population (manufacturing)	3808
Target population (services)	5032
Achieved sample	3587 (covering the census part of large enterprises, 360)
Achieved sample (manufacturing)	1671 (covering the census part of large enterprises, 204)
Achieved sample (services)	1916 (covering the census part of large enterprises, 156)
Weighted unit response rate	
Unweighted unit response rate	65,6
Definition of strata	Stratified simple random sampling. The variables used for the stratification were the economic activity of the enterprise (according to NACE Rev.2 classification) and the enterprise size class (10-49, 50-249, 250+ as a census part).
Weights calculation methods: Variables used and methodology	Two types of weights based on numbers of enterprises and turnover are used. Weights based on enterprise numbers for qualitative variables: N/n , number of enterprises in a stratum (in the frame)/number of responding enterprises (valid responses). Weights based on turnover for quantitative variables: total turnover of the enterprises in a stratum (in the frame)/total turnover of the responding enterprises (valid responses). Over-coverage and extreme values are accounted for in the calculations.
Implementation of non-response survey	-

Number and types of reminders	Three letter reminders, two email reminders/contacts and CATI telephone contacts and reminders
Use of imputation methods	Nominal scale variables by mode of strata, ordinal scale variables by median of strata, metric variables by proportional coefficients in strata
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	https://tilastokeskus.fi/til/inn/index_en.html
Survey scope includes public (i.e. central, regional or local government-controlled enterprises" (Yes/no/explain)	Yes

France

Name of survey	Eurostat Community Innovation Survey 2018 and National Innovation Survey
Name of responsible agency	Insee
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	May 2019 to December 2019
First release of innovation results	August 2020 (Insee web site)
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions. Specific examples are provided in the annex.
Differences with the OSLO manual	None
Difference with the CIS questionnaire	Mainly adopt the CIS common questionnaire as a model. Some Eurostat optional questions were removed, and we add some questions.
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73. Full coverage for the French survey: NACE Rev. 2 sections B to N (except divisions 75 and 82)
Sampling frame used	National statistical business register
Completion requirement	Compulsory
Unit of analysis surveyed	The legal unit (UL), identified by its SIREN number, with the exception of 12 units which are interrogated as profiled companies (EP).
Survey type	Combined A sample survey was used to collect data for the enterprises of less than 250 employees. Concerning the enterprises of 250 employees and more, they were covered by complete enumeration (census).
Type of interviews	Web and postal questionnaires
Average time to complete the survey	30 minutes
Target population	150 648 units (information for the full coverage of the French survey)
Target population (manufacturing)	
Target population (services)	
Achieved sample	20 150 units
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	71 %
Definition of strata	For the sampled enterprises stratified random sampling was applied. The variables used for the stratification of the sample were the economic activity of the enterprise (according to NACE Rev.2classification),the enterprise size (10-19, 20-49, 50-249, 250+), and the region where possible.
Weights calculation methods: Variables used and methodology	Correction of total non-response by reweighting using the homogeneous response group method, calibration on margins and winsorisation.
Implementation of non-response survey	No non-response survey.
Number and types of reminders	Exhaustive mail reminders on non-respondents, priority telephone reminders on non-respondents
Use of imputation methods	Yes (correction of partial non-response)
Combined with other surveys	No
Link to documentation (outputs and	https://www.insee.fr/fr/metadonnees/source/operation/s1477/pre

questionnaire)

sentation

Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)

Germany

Name of survey	Community Innovation Survey (CIS 2018)
Name of responsible agency	ZEW - Leibniz Centre for European Economic Research, on behalf of the Federal Ministry of Education and Research (BMBF)
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	February to July 2019
First release of innovation results	January 2020
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions. Specific examples are provided in a fold-out part of the questionnaire.
Differences with the OSLO manual	None
Difference with the CIS questionnaire	All CIS questions were included, following the harmonised CIS questionnaire.
Target population cut-off point	5 or more employees; for indicators delivered to OECD: 10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73. The national survey also includes M69, M70.2, M74, N78 to N82. The national survey also includes enterprises with 5 to <10 employed persons in all sectors covered by the survey.
Sampling frame used	Administrative sources (business register) for determining the size of the target population by stratum; commercial sources (Creditreform data base, containing about the same number of enterprises as the business register) for drawing the sample.
Completion requirement	Voluntary
Unit of analysis surveyed	Enterprise
Survey type	Combined (Combination of sample survey and complete enumeration (i.e. census) of the enterprises included in the frame population. The employment size class is used to define a threshold. Larger enterprises are enumerated, while smaller enterprises are sampled. The threshold to separate large from smaller enterprises for Germany is 500 employees).
Type of interviews	Web and postal questionnaire
Average time to complete the survey	about 1.5 hours (for CIS 2014, no more recent data available)
Target population	147,759 (for CIS; 299,638 for national reporting)
Target population (manufacturing)	68,985 (for CIS; 100,416 for national reporting)
Target population (services)	78,774 (for CIS; 199,222 for national reporting)
Achieved sample	5,785 (for CIS; 8,808 for national reporting) Non-response survey: 6,598 (for CIS; 10,330 for national reporting)
Achieved sample (manufacturing)	3,604 (for CIS; 4,498 for national reporting) Non-response survey: 4,322 (for CIS; 5,524 for national reporting)
Achieved sample (services)	2,181 (for CIS; 4,310 for national reporting) Non-response survey: 2,276 (for CIS; 4,806 for national reporting)
Weighted unit response rate	-
Unweighted unit response rate	23.9% (for CIS; 21.9% for national reporting) Response rate incl. participants of the non-response survey: 51.2% (for CIS; 49.7% for national reporting)
Definition of strata	For the sampled enterprises a stratified random sampling was applied. The variables used for the stratification of the sample were the economic activity of the enterprise (according to NACE Rev.2 classification) and the enterprise size (5-9, 10-19, 20-49, 50-99, 100-

	249, 250-499, 500-999, 1000+ employees). The stratified sample is disproportional according to innovation indicators such as innovation expenditures as a percentage of sales, share of innovative enterprises).
Weights calculation methods: Variables used and methodology	We use three weights: enterprise weight (for qualitative variables), sales weight (for expenditure and sales variables) and employee weight (for employment variables). The German CIS applies simple weighting (qualitative variables such as the number of innovators) and bounded weighting (quantitative variables such as innovation expenditure or sales with new products).
Implementation of non-response survey	Yes, a comprehensive non-response survey is carried out, based on a stratified sample of all non-responding firms. For each firm in the NR sample, up to three replacement firms are drawn, which are surveyed in case a sampled firm refused to participate in the NR survey. The NR survey is carried out by telephone and collects data on product innovation, process innovation, ongoing or abandoned innovation activities, and in-house R&D activities
Number and types of reminders	2 reminders per telephone followed by a letter and/or email containing the questionnaire plus link to the online version.
Use of imputation methods	We use longitudinal imputation based on firm information from prior surveys and cross-section imputation based on the mean of strata.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	www.zew.de/innovation
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises (Yes/no/explain)	Yes - all enterprises with economic activities in the target industries of the innovation survey form the target population, irrespective of ownership. The only exception is Nace sector 72. Here, public research organisations are excluded from the target population and are not surveyed. Firms with public ownership dominate in Nace sectors 36 and 37 and represent a significant share of all firms in the target population in Nace sectors 35 and 60.

Greece

Name of survey	Community Innovation Survey 2018
Name of responsible agency	National Documentation Centre (EKT)
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	November 2019 – January 2020
First release of innovation results	March 2020
Concepts, notes and definitions in the questionnaire	All relevant concepts, notes and definitions, as well as instructions for the completion of the questionnaire are included in the questionnaire, within the questions, in the form of explanatory text, pop-ups (for the online version), footnotes and/or hyperlinks, following the structure of the CIS 2018 harmonised survey questionnaire
Differences with the OSLO manual	None
Difference with the CIS questionnaire	The CIS 2018 harmonised survey questionnaire is fully adopted.
Target population cut-off point	10 or more persons employed
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	The national statistical Business Register of reference year 2018, maintained by the Hellenic Statistical Authority
Completion requirement	Mandatory
Unit of analysis surveyed	Enterprise
Survey type	Combination of census and sample survey: census for enterprises over 500 persons employed and R&D 2017 performers / sampling from the Statistical Business Register for enterprises in the remaining size classes of the target population
Type of interviews	Online survey supported by intensive telephone contacts and on-site visits by experienced interviewers
Average time to complete the survey	1h
Target population	12,213
Target population (manufacturing)	4,664
Target population (services)	7,170
Achieved sample	6,505
Achieved sample (manufacturing)	2,870
Achieved sample (services)	3,261
Weighted unit response rate	59.4%
Unweighted unit response rate	60.6%
Definition of strata	A one-stage stratified random sampling is applied for the selection of the CIS 2018 sample. The strata in the sampling frame are defined based on the following three variables: - 2-digit sector of economic activity (in accordance with NACE Rev.2 classification specifications) - Region (NUTS 2 level) - Enterprise size class according to the number of persons employed (10-19, 20-49, 50-99, 100-249, 250-499, 500+ persons employed)
Weights calculation methods: Variables used and methodology	Weights are calculated as the inverse of the sampling fraction, using the total number of enterprises in each stratum divided by the number of enterprises in the realised sample in the same stratum. By taking into account the realised sample, weights are adjusted for unit non-response.
Implementation of non-response survey	A non-response survey is carried out to a simple random sample of around 10% of non-responding units, given that the unit non-response

	rate exceeds 30%. The results of the non-response survey do not indicate significant differences between respondents and non-respondents, therefore, there is no need for further adjustment of the weighting factors.
Number and types of reminders	Three e-mail reminders were sent to the enterprises followed by intensive telephone contacts, personal e-mails and close follow-up by experienced interviewers.
Use of imputation methods	Imputation is applied on total turnover and on the intramural R&D expenditure for approximately 4.8% of the respondents. For turnover, the imputed figures are based on information available in the balance sheets and other administrative sources while for R&D expenditure, the missing information is completed with the respective figure from R&D survey. In both cases, the relevant sources are consulted only for those enterprises that do not provide the missing figures during follow-up.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	Documentation is currently available online only in Greek: https://metrics.ekt.gr/sites/metrics-ekt/files/pages-pdf/SIMS_%20CISstatistics_el_2018.pdf
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	No

Hungary

Name of survey	Eurostat Community Innovation Survey 2018 and National Innovation Survey
Name of responsible agency	Hungarian Central Statistical Office
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	September 2019 – January 2020
First release of innovation results	April 2020
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions
Differences with the OSLO manual	None
Difference with the CIS questionnaire	Model questionnaire fully adopted
Target population cut-off point	10 employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Compulsory, but not enforceable in practice
Unit of analysis surveyed	Enterprise
Survey type	Combined (Enterprises with less than 100 employees are part of the sample survey, enterprises more or equal 100 employees are enumerated completely)
Type of interviews	Use of electronic questionnaire
Average time to complete the survey	
Target population	16752
Target population (manufacturing)	8297
Target population (services)	8455
Achieved sample	7831
Achieved sample (manufacturing)	4743
Achieved sample (services)	3088
Weighted unit response rate	89.1%
Unweighted unit response rate	90.4%
Definition of strata	Stratification was made according to CIS methodological recommendations. The variables used for the stratification were the economic activity of the enterprise according to NACE rev.2 classification at the 2 digits (division) level, enterprise size (number of employees), and enterprise location by NUTS-2 region.
Weights calculation methods: Variables used and methodology	Variables: Number of enterprises Methodology: The basic method for adjusting for different probabilities of selection used in the sampling process is to use the inverse of the sampling fraction i.e. using the number of enterprises. Our method is based on the figure Nh/nh where Nh is the total number of enterprises in stratum h of the population and nh is the number of enterprises in the realised sample in stratum h of the population, assuming that each unit in the stratum had the same inclusion probability. This automatically adjusts the sample weights of the respondents to compensate for unit non-response.
Implementation of non-response survey	No non-response survey
Number and types of reminders	Two postal reminders and one or two phone calls
Use of imputation methods	No imputation was made
Combined with other surveys	No

Link to documentation (outputs and questionnaire)	https://www.ksh.hu/stadat_eng
Survey scope includes public (i.e. central, regional or local government-controlled enterprises" (Yes/no/explain)	Yes

Iceland

Name of survey	Community Innovation Survey 2018
Name of responsible agency	
Reference period	2016-2018
Length of reference period	3 years
Reference year	
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	
Differences with the OSLO manual	
Difference with the CIS questionnaire	
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	
Completion requirement	
Unit of analysis surveyed	Enterprise
Survey type	
Type of interviews	
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	
Number and types of reminders	
Use of imputation methods	
Combined with other surveys	
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	

Ireland

Name of survey	Community Innovation Survey 2018
Name of responsible agency	Central Statistics Office (CSO)
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions
Differences with the OSLO manual	None
Difference with the CIS questionnaire	
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Compulsory
Unit of analysis surveyed	Enterprise
Survey type	
Type of interviews	
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	No non-response survey used
Number and types of reminders	Four reminders including a Final reminder - issued as a letter.
Use of imputation methods	
Combined with other surveys	
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	

Name of survey	Eurostat Community Innovation Survey 2018 and National Innovation Survey
Name of responsible agency	ISTAT
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	October-December 2019
First release of innovation results	December 2020
Concepts, notes and definitions in the questionnaire	In footnotes and annex
Differences with the OSLO manual	None
Difference with the CIS questionnaire	It follows completely, the CIS harmonised structure proposed by Eurostat.
Target population cut-off point	10 or more persons employed
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Mandatory
Unit of analysis surveyed	Enterprise
Survey type	Combined (Combination of sample survey and complete enumeration (i.e. census) of the enterprises included in the frame population. The employment size class is used to define a threshold. Larger enterprises are enumerated, while smaller enterprises are sampled. The threshold to separate large from smaller enterprises is 250 persons employed)
Type of interviews	It's a Web survey. Use of online questionnaire
Average time to complete the survey	n.a.
Target population	Core population: 112,150
Target population (manufacturing)	70,116
Target population (services)	42,034
Achieved sample	19,359
Achieved sample (manufacturing)	11,100
Achieved sample (services)	8,259
Weighted unit response rate	70.1%
Unweighted unit response rate	70.1%
Definition of strata	The strata are defined by combining the economic activities (NACE classification), size classes (number of persons employed) and regions (NUTS classification) by the domains of interest.
Weights calculation methods: Variables used and methodology	Calibration estimators methodology were used for the estimation process [Deville, J.C. and Sarndal, C.E. (1992) Calibration estimators in survey sampling, Journal of the American Statistical Association 87, 367.382]. It can be applied to the extent that the known totals of some auxiliary variables, strictly correlated to the variables of interest, are available. The final weights are obtained by adopting the following procedure: an initial weight is assigned to each sampled unit with reference to the sampling plan as the reciprocal of the inclusion probability. Two correction factors for initial weights are then calculated: a first one is the unit non response factor; a second one is to satisfy equality between estimation of auxiliary variables and known totals from the Register. The final weights are thus obtained as the result of the product between initial weights and correction factors. The number of enterprises and the number of persons employed were used as auxiliary

	variables, according to the information provided by the Italian Official Business Register ASIA.
Implementation of non-response survey	None
Number and types of reminders	2 email reminders
Use of imputation methods	Yes. A first type of imputation was the logical (or deductive) imputation, that was applied to correct inconsistencies or for imputing some item non response, based on the answers provided in related questions. After having cleaned all the data, the imputation programs went on correcting metric variables separately from ordinal variables. As far as the metric variables are concerned, a weighted mean of each metric variable, by NACE and size class, was calculated (after having removed outliers) and applied as a ratio to the enterprises with the missing values within each stratum. The imputation of the ordinal and nominal was done after the metric estimation. The technique used is nearest-neighbor hot decking. Nearest-neighbor imputation uses data from clean records (free of logical inconsistency) to impute missing values of recipient records. The donors were chosen in such a way that the distance between the donor and the recipient is minimized. The distance measure is a multivariate measure based on the reported data.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	L'innovazione nelle imprese. Anni 2016-2018 L'innovazione nelle imprese (istat.it) Rilevazione statistica sull'innovazione nelle imprese (istat.it)
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	Yes

Japan

Name of survey	Japanese National Innovation Survey 2020 (J-NIS 2020)
Name of responsible agency	National Institute of Science and Technology Policy (NISTEP)
Reference period	2017–2019
Length of reference period	3 years
Reference year	2019
Data collection period	November 2020
First release of innovation results	October 2021
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions, in footnotes and in the annex
Differences with the OSLO manual	The concepts, definitions of terms, and methodology are based on those in the <i>Oslo Manual 2018</i> ; and some question items are also used on the basis of the recommendations in the <i>Oslo Manual 2018</i> . Non-response survey is not conducted in this wave.
Difference with the CIS questionnaire	The quantitative data in Q.3.10 (innovation expenditures, except for R&D expenditures (i.e. expenditures for R&D performed in-house and R&D contracted out to others)) were not collected.
Target population cut-off point	10 persons employed
Industry coverage	Classification: JSIC Rev.13 Core coverage: Enterprises in the JSIC Rev.13 divisions/major groups/group C, E, R90, F, R88, I50–55, H, Q86, G, N801, J, L71, and L73–74, which correspond to the ISIC Rev.4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71–73. At the industrial level, there are some gaps in the following correspondences between the JSIC Rev.13 and the ISIC Rev.4/NACE Rev.2: E27–28 and E30 to C26; E25–26 to C28; E13, E32 and R90 to C31–33; F36 to E36; R88 to E37–39, and I50–55 to G46. Non-core coverage: Enterprises in the JSIC Rev.13 divisions/major groups/group A, B, D, R89, I56–61, M75–77, K68–70, L72, R91–92, and N791, which correspond to the ISIC Rev.4/NACE Rev.2 sections/divisions A, F, G45, G47, I, L, N77, M69–70, M74, N78, N80–82, and N79.
Sampling frame used	National statistical business register
Completion requirement	Voluntary
Unit of analysis surveyed	Enterprise which has the legal status of <i>kabusiki-kaisha</i> (stock company) (including former <i>yūgen-kaisha</i> (limited liability company)), <i>gōmei-kaisha</i> (general partnership company), <i>gōshi-kaisha</i> (limited (liability) partnership company), <i>gōdō-kaisha</i> (limited liability company), or <i>sōgo-kaisha</i> (mutual company).
Survey type	Combined (Sample survey for the strata of the enterprises with 10–49, 50–249, and 250–999 persons employed; Census for the strata of the enterprises with 1,000–4,999 and 5,000+ persons employed)
Type of interviews	Web and postal questionnaire
Average time to complete the survey	(Not available)
Target population	Core coverage for the OECD Innovation Indicators: 224,238 (Total including non-core coverage: 442,978)
Target population (manufacturing)	107,108
Target population (services)	Core coverage: 109,867 (Total including non-core coverage: 251,461)

Achieved sample	Core coverage: 9,136 (Total including non-core coverage: 12,534)
Achieved sample (manufacturing)	5,834
Achieved sample (services)	Core coverage: 2,726 (Total including non-core coverage: 5,341)
Weighted unit response rate	Core coverage: 44% (Total including non-core coverage: 43%)
Unweighted unit response rate	Core coverage: 41% (Total including non-core coverage: 40%)
Definition of strata	For the sampled enterprises, a stratified random sampling was applied. The variables used for the stratification of the sample were the economic activity of the enterprise (according to JSIC Rev.13 classification) and the enterprise size (10–49, 50–249, 250–999, 1,000–4,999, 5,000+ persons employed).
Weights calculation methods: Variables used and methodology	Weights were calculated as inverses of the ratios of the numbers of enterprises in the achieved sample to the numbers of enterprises in the target population by sampling stratum.
Implementation of non-response survey	No
Number and types of reminders	At least one postal reminder and one telephone reminder
Use of imputation methods	Yes
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	Statistical reports (in Japanese): https://doi.org/10.15108/nr192 Statistical tables (in Japanese): https://www.e-stat.go.jp/stat-search/files?page=1&layout=datalist&toukei=00400503&tstat=000001039433&cycle=0&tclass1=000001158973&tclass2val=0 Questionnaire: https://www.nistep.go.jp/wp/wp-content/uploads/7b60d14bc7590266526b1cdae632dc4c.pdf Questionnaire (English translation): https://www.nistep.go.jp/wp/wp-content/uploads/J-NIS2020_Questionnaire_en_sample.pdf
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises” (Yes/no/explain)	Public enterprises which had the legal status of <i>kabusiki-kaisha</i> (stock company) (including former <i>yūgen-kaisha</i> (limited liability company)), <i>gōmei-kaisha</i> (general partnership company), <i>gōshi-kaisha</i> (limited (liability) partnership company), <i>gōdō-kaisha</i> (limited liability company), or <i>sōgo-kaisha</i> (mutual company) were covered.

Korea

Name of survey	Korean Innovation Survey (KIS)
Name of responsible agency	Science and Technology Policy Institute (STEPI)
Reference period	2017-19/2018-20
Length of reference period	3 years
Reference year	
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions. In the footnotes and annex.
Differences with the OSLO manual	None
Difference with the CIS questionnaire	All CIS questions are included, but a few questions are modified (merged) or added based on the Korean context.
Target population cut-off point	10 or more employees
Industry coverage	
Sampling frame used	'Census of Establishment' by the Statistics Korea (KOSTAT)
Completion requirement	Voluntary
Unit of analysis surveyed	Enterprise
Survey type	Sample survey
Type of interviews	Combined (web, electronic, FAX, visit to enterprises)
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	N/A
Unweighted unit response rate	
Definition of strata	Type of business and number of workers
Weights calculation methods: Variables used and methodology	Variables: Number of Employees (10-49, 50-99, 100-299, 300-499, above 500) Industry Sectors: 56 sectors (Manufacturing 23 sectors, Services 33 sectors) Methodology: Neyman allocation method and cut-off sampling (above 500)
Implementation of non-response survey	Unit non-response: Re-contacting the enterprises and sample substitution (abiding by the applied rule of stratification) Item non-response: non-response survey
Number and types of reminders	At least 3 times, up to 7 times by telephone
Use of imputation methods	N/A
Combined with other surveys	N/A
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled enterprises" (Yes/no/explain)	

Latvia

Name of survey	Community Innovation Survey 2018
Name of responsible agency	Central Statistical Bureau of Latvia
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	February 2019 – July 2019
First release of innovation results	July 2020
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions
Differences with the OSLO manual	None
Difference with the CIS questionnaire	Questionnaire includes one additional question for national needs.
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National Statistical Business Register
Completion requirement	Compulsory, with enforceable penalties
Unit of analysis surveyed	Enterprise
Survey type	Data were collected through a combination of both census and sampling.
Type of interviews	Web questionnaire and phone assisted interviews
Average time to complete the survey	90 minutes
Target population	4736
Target population (manufacturing)	2017
Target population (services)	2719
Achieved sample	3103
Achieved sample (manufacturing)	1367
Achieved sample (services)	1736
Weighted unit response rate	91.5
Unweighted unit response rate	92.1
Definition of strata	The method adopted was stratified simple random sampling. Enterprises were stratified by the size class of enterprise and economic activity by NACE 2.red. classifications.
Weights calculation methods: Variables used and methodology	The inverse of the sampling fractions was used as weights. All large statistical units had weights=1. In the realized sample weights= N_h/m_h where N_h is the total number of enterprises in the stratum h of the population and m_h is the number of enterprises in the realised sample in the stratum h , assuming that each unit in the stratum had the same inclusion probability. This will automatically adjust the sample weights of the respondents to compensate for unit non-response.
Implementation of non-response survey	The non-response survey was not conducted
Number and types of reminders	Approximately 4 e-mails and 4 calls
Use of imputation methods	No imputations were made
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	https://stat.gov.lv/en/statistics-themes/business/innovations
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	The survey scope includes public enterprises.

Lithuania

Name of survey	Community Innovation Survey 2018
Name of responsible agency	Lithuanian Department of Statistics
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	September 2019 - October 2019
First release of innovation results	December 2019
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions.
Differences with the OSLO manual	No differences
Difference with the CIS questionnaire	Practically identical. Not all voluntary questions are included.
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, F, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Compulsory
Unit of analysis surveyed	Enterprise
Survey type	Combined (Type of survey for enterprises with 250 and more employees is complete enumeration. For enterprises with 10-49 and 50-249 employee is a sample survey).
Type of interviews	Electronic questionnaires with a possibility to fill online or print and fill in a paper version
Average time to complete the survey	The average time spent by a respondent on the filling in of the statistical questionnaire was 78 min.
Target population	9972 enterprises
Target population (manufacturing)	5334 enterprises
Target population (services)	4638 enterprises
Achieved sample	2433 enterprises
Achieved sample (manufacturing)	1349 enterprises
Achieved sample (services)	1084 enterprises
Weighted unit response rate	Core NACE (B-C-D-E-46-H-J-K-71-72-73) – Weighted unit response rate – 98.85 per cent
Unweighted unit response rate	Core NACE (B-C-D-E-46-H-J-K-71-72-73) – Un-weighted unit response rate – 99.01 per cent
Definition of strata	The variables used for the stratification of the sample were the economic activity of the enterprise (according to NACE Rev.2 classification, 2 digits) and the enterprise size (10-49, 50-249, 250+). All enterprises from small groups (less than 15 units) by NACE Rev 2 activities and number of persons employed groups were included into the sample. The enterprises with 250+ employees are covered by complete enumeration. Within every NACE Rev 2 group the strata of enterprises were formed by number of persons employed groups. Then the Neyman optimal allocation (with variable number of persons employed) was used for determination of the sample size for each stratum specified. In each stratum simple random sampling was used.
Weights calculation methods: Variables used and methodology	The variables used for weighting were the same as for stratification, i. e. the economic activities and enterprise size according to the number of employees. Methodology: There were some divergences between the target population and the frame population due of economically not active enterprises. These errors were estimated according to the

	sample design. Economically non active enterprises were excluded from target population. All results were weighted against the total population by means of sample weights. The weights were established by using the probability of each enterprise to be included in the sample. The primary strata were not changed and the primary weights were used for actual estimation domains. The calibration method was not used.
Implementation of non-response survey	No non-response survey was carried out.
Number and types of reminders	2 automatic reminders via e. system, later 3 emails or calls were made before enterprises were coded as non-responding.
Use of imputation methods	No imputation was made.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	Methodological documents are published in the Official Statistics Portal section Innovation activities. Statistical indicators are published in the Database of Indicators (Science and technology -> Innovation activities). For the collection of statistical data a statistical report form has been prepared - the statistical report on innovation activities INV-01 (every 2 years). The statistical report form is published at the following address: http://estatistika.stat.gov.lt/statistiniu-ataskaitu-formos.html (only in Lithuanian).
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	Yes

Luxembourg

Name of survey	Community Innovation Survey 2018
Name of responsible agency	
Reference period	2016-2018
Length of reference period	3 years
Reference year	
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	
Differences with the OSLO manual	
Difference with the CIS questionnaire	
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	
Completion requirement	
Unit of analysis surveyed	Enterprise
Survey type	
Type of interviews	
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	
Number and types of reminders	
Use of imputation methods	
Combined with other surveys	
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	

Netherlands

Name of survey	Community Innovation Survey 2018
Name of responsible agency	Statistics Netherlands – CBS
Reference period	2016-2018
Length of reference period	3 years
Reference year	
Data collection period	2019
First release of innovation results	2020
Concepts, notes and definitions in the questionnaire	Included within or immediately following questions. Details are also provided in the footnotes and in the annex.
Differences with the OSLO manual	None.
Difference with the CIS questionnaire	Only mandatory questions are included.
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register.
Completion requirement	Voluntary.
Unit of analysis surveyed	Enterprise.
Survey type	Sample survey, although for the higher size classes the sampling probability was 1 for most NACE.
Type of interviews	Online questionnaire.
Average time to complete the survey	N/A.
Target population	27099
Target population (manufacturing)	8771
Target population (services)	18328
Achieved sample	7384
Achieved sample (manufacturing)	2673
Achieved sample (services)	4711
Weighted unit response rate	24%
Unweighted unit response rate	20%
Definition of strata	Sampling by NACE x size class.
Weights calculation methods: Variables used and methodology	Number of enterprises.
Implementation of non-response survey	The non-response survey was not conducted.
Number and types of reminders	Two reminders (paper letters by traditional mailing) are sent out to all enterprises in the sample that have not responded. In some cases after these two recalls some vital enterprises are contacted via telephone in order to convince them to respond after all.
Use of imputation methods	None.
Combined with other surveys	Yes (R&D survey).
Link to documentation (outputs and questionnaire)	Not available.
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises” (Yes/no/explain)	No.

New Zealand

Name of survey	Business Operation Survey 2019
Name of responsible agency	Stats NZ
Reference period	2017-2018
Length of reference period	2 years
Reference year	2018
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions
Differences with the OSLO manual	
Difference with the CIS questionnaire	The Business Operations Survey Innovation Module follows closely with the CIS questionnaire, as it was used as a model for the development of the Innovation Module. There are some differences in ordering, and some questions from the CIS are not present in the Business Operations Survey. These differences are due to the need to make the questionnaire more applicable and easy to fill in for New Zealand businesses. The differences are also due to stakeholder needs for only certain types of information. With this in mind, the Business Operations Survey follows the CIS model as closely as possible, but with some adjustments to suit to the New Zealand environment.
Target population cut-off point	
Industry coverage	The sectoral coverage of enterprises matches the CIS scope, except for the "Wholesale trade, except of motor vehicles and motorcycles" G46 section which excludes "motor vehicle and motor vehicle parts wholesaling".
Sampling frame used	National statistical business register
Completion requirement	Compulsory, but not enforceable in practice
Unit of analysis surveyed	Enterprise
Survey type	Sample survey
Type of interviews	Postal questionnaire
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	Once the target response rate has been met, there is no follow up of any remaining non responders. We account for this non-response by weighting up the responding enterprises. This is the same method used for any enterprises that have ceased trading.
Number and types of reminders	Up to three postal or email reminders are used for this survey. If an enterprise has not responded after the three postal or email reminders, then they are contacted directly via phone.
Use of imputation methods	
Combined with other surveys	Yes
Link to documentation (outputs and questionnaire)	

Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)

Norway

Name of survey	Community Innovation Survey 2018
Name of responsible agency	
Reference period	2016-2018
Length of reference period	3 years
Reference year	
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	
Differences with the OSLO manual	
Difference with the CIS questionnaire	
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	
Completion requirement	
Unit of analysis surveyed	Enterprise
Survey type	
Type of interviews	
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	
Number and types of reminders	
Use of imputation methods	
Combined with other surveys	
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	

Poland

Name of survey	Survey on Innovation in industry in 2016-2018/Survey on innovation in the service sector in 2016-2018 (2 surveys)
Name of responsible agency	Statistics Poland
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	February – June 2019
First release of innovation results	28 th October 2019
Concepts, notes and definitions in the questionnaire	Some included within questions and in footnotes. Details are provided in the annexed explanation notes.
Differences with the OSLO manual	Generally no differences, persons employed instead of employees
Difference with the CIS questionnaire	Additional questions for national needs, persons employed instead of employees
Target population cut-off point	10 or more persons employed
Industry coverage	Classification: PKD 2007 (Polish Classification of Activities coherent with NACE Rev. 2) Sections/Divisions B, C, D, E, G46, H, J, K, M71-72-73
Sampling frame used	National statistical business register
Completion requirement	Mandatory
Unit of analysis surveyed	Legal unit
Survey type	Combination of sample survey and census
Type of interviews	On-line questionnaire
Average time to complete the survey	197 minutes (including time for preparing data)
Target population	64080
Target population (manufacturing)	33757
Target population (services)	30323
Achieved sample	21286
Achieved sample (manufacturing)	14988
Achieved sample (services)	6298
Weighted unit response rate	68.2%
Unweighted unit response rate	78.0%
Definition of strata	Strata were defined as intersections of PKD (NACE) divisions, geographical regions (NUTS 2 level – voivodships) and enterprise sizes (10-49 persons employed, 50-249, 250+)
Weights calculation methods: Variables used and methodology	The base weights were calculated separately for strata as ratios: the number of enterprises of the frame population to the number of enterprises in the sample. The base weights were corrected for unit non-response by multiplying the base weight by adjustment factor. The non-response adjustment factors were computed using information about reasons of non-response (e.g. unit was inactive or there was incorrect contact information).
Implementation of non-response survey	No
Number and types of reminders	3 e-mail reminders for all non-respondents and additionally urging notes, e-mails and phone calls for still non-respondents
Use of imputation methods	Hot-deck
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	Questionnaires: http://form.stat.gov.pl/formularze/2019/passive/PNT-02.pdf http://form.stat.gov.pl/formularze/2019/passive/PNT-02u.pdf results: https://stat.gov.pl/en/topics/science-and-technology/science-and-technology/innovative-activity-of-enterprises-in-the-years-20162018,3,4.html https://bdl.stat.gov.pl/BDL/start

Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)

Yes, but only enterprises, not any other organisational forms

Portugal

Name of survey	Community Innovation Survey 2018
Name of responsible agency	DGEEC/MCTES (Directorate General of Education and Science Statistics/Portuguese Ministry of Science, Technology and Higher Education) in collaboration with national statistics office (Statistics Portugal)
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	May 2021-November 2021
First release of innovation results	April 2021
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions. The annex and footnotes provide general details and examples.
Differences with the OSLO manual	None
Difference with the CIS questionnaire	None
Target population cut-off point	10 or more persons employed
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Compulsory
Unit of analysis surveyed	The enterprise as legal unit.
Survey type	Combined (Portuguese Community Innovation Survey was collected using a combination of methods census and sample survey. Census operation in the case of large enterprises (i.e., with ≥ 250 employees) and a sample survey for the other strata).
Type of interviews	Electronic questionnaire.
Average time to complete the survey	72 minutes
Target population	20737
Target population (manufacturing)	12383
Target population (services)	8354
Achieved sample	7786
Achieved sample (manufacturing)	4626
Achieved sample (services)	3160
Weighted unit response rate	91,2%
Unweighted unit response rate	92,8%
Definition of strata	The sampling frame was stratified by economic activity, enterprise size, according to the persons employed (10-49, 50-249 and 250+) and NUTS2 region. The sample size was determined in order to guarantee the recommended level of precision.
Weights calculation methods: Variables used and methodology	Variables: The variables used for weighting are all three stratification variables (the economic activities in accordance with NACE, the size-classes and Regional aspects at NUTS 2 level). Methodology: For the sample weights we used the inverse of the sampling fraction (using the number of enterprises). No calibration method was used.
Implementation of non-response survey	It wasn't necessary to implement a non-response survey.
Number and types of reminders	There are usually 7 reminders, an initial reminder of the response deadline, followed by 6 reminders to enterprises that have not responded after the initial reminder. Most reminders are sent by email. For enterprises that are not registered on the collection platform, reminders are sent by postal mail.
Use of imputation methods	All the questions of the PT CIS survey are mandatory and the electronic platform does not allow the submission of the

	questionnaire if there are unanswered questions. Inconsistent and outlier values are checked with the enterprises and if necessary corrections are made. The information about turnover and number of persons employed are directly imputed from the statistical business register.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	https://www.dgeec.mec.pt/np4/207/ https://smi.ine.pt/SuporteRecolha/Detalhes/?id=10354&lang=PT
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	Yes

Slovak Republic

Name of survey	Eurostat Community Innovation Survey 2018 and National Innovation Survey
Name of responsible agency	The Statistical Office of the Slovak Republic
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	May – September 2019
First release of innovation results	06/2020
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions, in the footnotes or in the annex
Differences with the OSLO manual	None
Difference with the CIS questionnaire	All mandatory questions and several voluntary questions from the Harmonised Questionnaire except questions 2.4, 2.8, 2.10, 2.11, 3.11, 3.17, 4.8 and 4.9 were included in national questionnaire.
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73. Non-core coverage: F
Sampling frame used	National statistical business register
Completion requirement	Compulsory, with enforceable penalties
Unit of analysis surveyed	Enterprise
Survey type	Combined (Combination of sample survey and complete enumeration (i.e. census) of the enterprises included in the frame population. The employment size class is used to define a threshold. Large enterprises are enumerated, while smaller enterprises are sampled. The threshold to separate large from smaller enterprises is 250 employees).
Type of interviews	Electronic questionnaire
Average time to complete the survey	2,62 hours
Target population	8499 (Core)
Target population (manufacturing)	3879
Target population (services)	4264
Achieved sample	2816 (Core)
Achieved sample (manufacturing)	1342
Achieved sample (services)	1315
Weighted unit response rate	83,5 % (Core)
Unweighted unit response rate	84,3 % (Core)
Definition of strata	For the sampled enterprises a stratified random sampling was applied. The variables used for the stratification of the sample were the economic activity of the enterprise (according to NACE Rev.2 classification 2 digit level) and the enterprise size (10-49, 50-249, 250+).
Weights calculation methods: Variables used and methodology	The initial weights were calculated as N_h/n_h , where N_h was the total number of enterprises in stratum h of the population and n_h was the number of enterprises in the sample in stratum h , assuming that each unit in the stratum had the same inclusion probability. Initial weights were updated after collection of questionnaires. As a last step, calibration was applied, using Calif tool in R. The variables used for calibration were the total number of enterprises, turnover and number of employed persons, all of them taken from the SBS.
Implementation of non-response survey	No
Number and types of reminders	Generally, number of maximum reminders were 2 phone calls. In several cases more than 2 phone calls were conducted.

Use of imputation methods	No
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	https://slovak.statistics.sk/wps/portal/ext/themes/multi/science/publications https://slovak.statistics.sk/wps/portal/ext/metadata/surveys/
Survey scope includes public (i.e. central, regional or local government-controlled enterprises" (Yes/no/explain)	No

Slovenia

Name of survey	Community Innovation Survey 2018
Name of responsible agency	
Reference period	2016-2018
Length of reference period	3 years
Reference year	
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	
Differences with the OSLO manual	
Difference with the CIS questionnaire	
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	
Completion requirement	
Unit of analysis surveyed	Enterprise
Survey type	
Type of interviews	
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	
Number and types of reminders	
Use of imputation methods	
Combined with other surveys	
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	

South Africa

Name of survey	South African Business Innovation Survey
Name of responsible agency	Centre for science, technology & innovation indicators
Reference period	2014-2016
Length of reference period	3 years
Reference year	2016
Data collection period	August 2017 to August 2019
First release of innovation results	28 August 2020

Concepts, notes and definitions in the questionnaire

An innovation is the introduction of a new or significantly improved product, process, organisational method, or marketing method by your enterprise. The innovation must be new to your enterprise, although it could have been originally developed by other enterprises.

A product innovation is the introduction to market of a new or significantly improved good or service with respect to its capabilities, such as improved user-friendliness, components, software or sub-systems. The innovation must be new to your enterprise, but it does not need to be new to your industry sector or market. It does not matter if the innovation was originally developed by your enterprise or by other enterprises. A good is usually a tangible object such as a smart phone, furniture, or packaged software, but downloadable software, music and film are also goods. A service is usually intangible, such as retailing, insurance, educational courses, air travel, consulting, etc. Please note: The latest terminology classifies “products” as consisting of both “goods” and “services”. For example a firm in the financial services sector may report a “new financial product”. The provision of innovative services is of increasing importance in competitive economies and the survey covers both manufacturing and services orientated firms.

Process innovation is the use of new or significantly improved methods for the production or supply of goods and services. The innovation (new or improved) must be new to your enterprise, but it does not need to be new to your industry sector or market. It does not matter if the innovation was originally developed by your enterprise or by other enterprises. Exclude purely organisational innovations such as changes in firm structure or management practice.

An organisational innovation is the implementation of new or significant changes in firm structure, business practice or management methods that are intended to improve your firm’s use of knowledge, the quality of your goods and services, or the efficiency of work flows. Exclude mergers or acquisitions, even if for the first time.

A marketing innovation is the implementation of new or significantly improved marketing or sales methods to increase the appeal of your goods and services or to enter new markets. Examples include changes in product design or packaging, product placement, product promotion or pricing BUT excludes seasonal, regular and other routine changes in marketing methods.

	A marketing innovation is the implementation of new or significantly improved marketing or sales methods to increase the appeal of your goods and services or to enter new markets. Examples include changes in product design or packaging, product placement, product promotion or pricing BUT excludes seasonal, regular and other routine changes in marketing methods. Innovation activities include the acquisition of machinery, equipment, software, licenses, engineering and development work, training, marketing and research and experimental development (R&D) when they are specifically undertaken to develop and/or implement a product or process innovation. The public sector includes government owned organisations such as local, regional and national administrations and agencies, schools, hospitals, and government providers of services such as security, transport, housing, energy, etc.
Differences with the OSLO manual	The survey followed the guidelines in the 3rd edition of the Oslo Manual (OECD, 2005). The survey design is informed by the structure of the national business register of Statistics South Africa (Stats SA), which is used to draw a suitable stratified random sample. The Oslo Manual recommends size cut-offs that are based on employment, including only businesses with ten or more employees. The Stats SA business register has insufficient information on employment, and hence the size classes are of necessity based on turnover.
Difference with the CIS quest.	The survey used the methodological recommendations for the Community Innovation Survey (CIS) series (CIS 2012) of the European Union (EU) countries, as provided by Eurostat, the Statistical Office of the European Commission. Indicators that are both relevant for South Africa and internationally comparable are produced using these guidelines
Target population cut-off point	The Oslo Manual recommends size cut-offs that are based on employment, including only businesses with ten or more employees. The Stats SA business register has insufficient information on employment, and hence the size classes are of necessity based on turnover.
Industry coverage	The sampling frame for the innovation survey is characterised by 6 main sectors, which were further subdivided into a number of subsectors. The division into subsectors was as follows: Sector 2 (Mining and quarrying: 6 substrata), Sector 3 (Manufacturing: 10 substrata), Sector 4 (Electricity, gas and water supply: 2 substrata), Sector 6 (Wholesale trade and retail trade: 2 subsectors), Sector 7 (Transport, storage and communications: 5 subsectors) and Sector 8 (Financial intermediation, computer & related activities, R&D, architectural & engineering activities and technical testing & analysis: 5 subsectors). Additionally, each of the strata is divided into four distinct size classes (large, medium, small and very small) based on turnover cut-offs..
Sampling frame used	National business register of Statistics South Africa (Stats SA)
Completion requirement	All/most questions
Unit of analysis surveyed	Enterprise or firm
Survey type	Cross-sectional, with original sample including all large enterprises
Type of interviews	Postal paper questionnaires with telephonic and email follow ups, telephonic interviews, emailed pdf questionnaires with telephonic and email follow ups, face-to-face interviews.
Average time to complete the survey	About 20 minutes

Target population	41 535
Target population (manufacturing)	15 064
Target population (services)	26 018
Achieved sample	642
Achieved sample (manufacturing)	259
Achieved sample (services)	336
Weighted unit response rate	Not calculated
Unweighted unit response rate	15.0%
Definition of strata	Each of the strata defined under Industry Coverage above is divided into four distinct size classes. Therefore, the frame consisted of $(6+10+2+2+5+5) \times 4 = 30 \times 4 = 120$ strata by sector and size.

Weights calculation methods: Variables used and methodology

Reduced Population

The presence of non-valid companies in the sample served as a basis for reducing the original population size to a proportion pN = valid population. An important assumption here is that the original population consisted of a number of untraceable or expired companies that are no longer part of the base population. The procedures make no allowance for growth in the population size in the form of new entrants. Resultant estimates will on balance tend to be conservative and a “good” guess of baseline levels of innovation. Valid population sizes were therefore calculated from the proportion of valid firms as ascertained from the fieldwork in the sampling process.

Non-response

Of the total (valid) sample in each substratum, a proportion, q responded to the survey. So we infer a population of respondents in each substratum of $R=qpN$, and a corresponding population of non-respondents, $NR=(1-q)pN$.

Innovators and non-innovators

Of the valid respondents, a proportion, k , were found to be innovative, so that the number of innovative firms among the R responders is $I1 = kR = kqpN$. In order to estimate the innovation rate among non-responders, a non-response survey was conducted in the form of a simple random sample (SRS). The SRS revealed a 72.1% innovation rate amongst the total NR . Therefore the total number of virtual innovators among the non-responders is $I2 = NR \times 72.1\%$. A vital assumption is made whereby the 72.1% rate of innovation was assumed amongst the estimated number of non-respondents in all strata. Within each of the 120 substrata, the total estimated population of innovators is $I = I1 + I2$. Implicitly we therefore have a number of non-innovators, $NI = pN - I$, in each substratum.

Missing weights

There were a number of substrata for which no estimate, k was calculable since no responses were returned for these strata. The problem of zero weight principally arose in the four sectors that contained subsectors. The consequence is that all the valid firms, R , in those strata were implicitly assumed to be non-innovators. In the light of the fact that similar substrata in the main sector and size class did actually contain a portion of innovators, there was a situation of potential undercounting of innovators overall (and a subsequent over-counting of non-innovators, and therefore overstating the true value of NI).

Revised weights

One option for revising the weights would be to assume the 72.1% from the non-response survey to apply to the valid entities in the “weightless” substrata, and enter $pN \times 72.1\%$ for those substrata. A more internally consistent method is to inflate the (weighted) substrata in the main sector and same size class by the proportion missing/“weightless” for that size and sector class. In this way some of the homogeneity of size and sector class is maintained. We then calculate I^* and NI^* , the inflated (where applicable) number of innovators and non-innovators respectively.

Final weights

For each substratum, two weights are then calculated. The first weight, $W1$ is equal to $I^* / n1$, where $n1$ is the number of innovators in the sample. Defined in this way, $W1$ will have to be zero for strata where $n1 = 0$ and $n2 > 0$. In the situation where both $n1$ and $n2$ equal zero, then the inflation factor kicks in and we inflate remaining categories in the sector. In this manner, any estimates of population totals inferred from the Innovator data will be conservative, and for certain substrata, no totals will emerge. Similar logic applies to the calculation of the second weight, $W2$, for non-innovators, which is equal to $NI^* / n2$, where $n2$ is the number of non-innovators in the sample. In the situation whereby the sample of innovators or non-innovators is small and sparsely scattered throughout the 120 substrata, we invoke the use of “weight-totalling”.

Implementation of non-response survey

A simple random sample non-response survey was conducted, as recommended by the Oslo Manual (OECD, 2005) for surveys that achieve response rates of less than 70%. The purpose of the non-response survey was to, as described in the weights calculation methodology above, correct for any bias that might arise due to businesses that did not respond to the survey being less or more innovative than those businesses that did respond. The non-response survey covered 493 or 15% of the businesses that did not respond to the main survey, and a response rate of 68.3% was achieved. The correction for bias due to non-response was implemented by adjusting the probability weights used to project the sample results to the target population of enterprises. This methodology also adjusts the weights for invalid enterprises (enterprises that were found to have merged or been liquidated).

Number and types of reminders

At least two telephonic and at least two email reminders

Use of imputation methods

None

Combined with other surveys

No

Link to documentation (outputs and questionnaire)

<http://curation.hsrc.ac.za/Datasets-KDCAAA.phtml>

Survey scope includes public (i.e. central, regional or local government-controlled) enterprises” (Yes/no/explain)

National population of businesses in the Statistics South Africa (Stats SA) business registrar

Spain

Name of survey	Eurostat Community Innovation Survey 2018 and National Innovation Survey
Name of responsible agency	National Statistics Office – INE
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	July 2019 – October 2019
First release of innovation results	December 2019
Concepts, notes and definitions in the questionnaire	Every question has a description below. More detailed definitions and examples can be found in the annex attached at the end of the questionnaire.
Differences with the OSLO manual	None
Difference with the CIS questionnaire	Most of the CIS questions are included. Questions about the enterprise's economic information are included in the first page of our questionnaire.
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register
Completion requirement	Obligatory
Unit of analysis surveyed	Enterprise
Survey type	Combined (Combination of sample survey and complete enumeration (i.e.census) of the enterprises included in the frame population. The employment size class is used to define a threshold. Larger enterprises are enumerated, while smaller enterprises are sampled. The threshold to separate large from smaller enterprises for Spain is 200 employees).
Type of interviews	Web and postal questionnaires with the possibility of telephone interview.
Average time to complete the survey	Not available
Target population	72,730
Target population (manufacturing)	33,655
Target population (services)	39,075
Achieved sample	25,859
Achieved sample (manufacturing)	14,754
Achieved sample (services)	11,105
Weighted unit response rate	14.62%
Unweighted unit response rate	12.15%
Definition of strata	A stratified random sampling was applied. The variables used for the stratification of the sample were the economic activity of the enterprise according to NACE Rev.2 classification (59 strata), enterprise size (10-49, 50-249, 250+: 3 strata) and NUTS 2-region (19 strata).
Weights calculation methods: Variables used and methodology	Weighting was performed by "number of enterprises". The weighting process was performed in two steps. Firstly, basic weights were calculated in a simple way by Nh/nh using the inverse of the sampling fraction. In a second step the weights were re-adjusted according to the results of the non-response analysis. The re-adjusted weights were the only ones used for producing the final results
Implementation of non-response survey	No non-response survey is carried out.

Number and types of reminders	3 postal reminders
Use of imputation methods	Some partial non-response is imputed. Also, since collection of innovation data is performed in a combined R&D and innovation survey, some total non-response is also imputed to cover the total non-response of the R&D survey.
Combined with other surveys	Yes (R&D)
Link to documentation (outputs and questionnaire)	https://www.ine.es/dyngs/INEbase/en/operacion.htm?c=Estadistica_C&cid=1254736176755&menu=metodologia&idp=1254735576669
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	Yes

Sweden

Name of survey	Eurostat Community Innovation Survey 2018 and National Innovation Survey
Name of responsible agency	Statistics Sweden
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	March-August 2019
First release of innovation results	2019-11-12
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions. The annex and footnotes provide general details and examples.
Differences with the OSLO manual	None
Difference with the CIS questionnaire	None
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	National statistical business register. All research institutes (regardless of size) are also included in the sample.
Completion requirement	Compulsory, but not enforceable in practice
Unit of analysis surveyed	Enterprise
Survey type	Combined (Combination of sample survey and complete enumeration of the enterprises included in the frame population. The employment size class is used to define a threshold. Usually larger enterprises are enumerated, while smaller enterprises are sampled. The threshold to separate large from smaller enterprises is 200 employees).
Type of interviews	Use of web-questionnaire
Average time to complete the survey	
Target population	21694
Target population (manufacturing)	6675
Target population (services)	15019
Achieved sample	6810
Achieved sample (manufacturing)	3121
Achieved sample (services)	3689
Weighted unit response rate	
Unweighted unit response rate	13.6%
Definition of strata	For the sampled enterprises a stratified random sampling was applied. The variables used for the stratification of the sample were the economic activity of the enterprise (according to NACE Rev.2 classification, 2 digits) and the enterprise size (10-49, 50-249, 250+).
Weights calculation methods: Variables used and methodology	The weight is calculated as the total number of enterprises in the population per stratum divided by the number of enterprises answering the questionnaire per stratum: N_h/n_h . N_h is the total number of enterprises in stratum h of the population. n_h is the number of enterprises, that answered the survey, in the sample in stratum h of the population. Assuming that each unit in the stratum had the same inclusion probability.
Implementation of non-response survey	None
Number and types of reminders	Two written reminders have been sent out.
Use of imputation methods	According to Eurostat's methodology and by use of the SAS tools developed by Eurostat.
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	

Survey scope includes public (i.e. central, regional or local government-controlled enterprises" (Yes/no/explain)	No
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Switzerland

Name of survey	Survey on Innovation Activities in the Swiss economy
Name of responsible agency	KOF-ETH, Zurich (commissioned by State Secretariat for Education, Research and Innovation, SERI)
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the quest.	Included within the questions
Differences with the OSLO manual	
Difference with the CIS questionnaire	For product or Business process innovation-active firms, data excludes methods for information processing or communication; business practices for organising procedures or external relations; methods of organising work responsibility, decision making or human resource management; and, marketing methods for promotion, packaging, pricing, product placement or after sales services.
Target population cut-off point	
Industry coverage	
Sampling frame used	National business register from the Federal Statistical Office
Completion requirement	Voluntary
Unit of analysis surveyed	Enterprise
Survey type	Combined (Full coverage for large firms - where the threshold for "large" varies between activity classes -; Stratified random sampling for others).
Type of interviews	Postal questionnaire
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	
Number and types of reminders	One, telephonically.
Use of imputation methods	
Combined with other surveys	Yes (ICT)
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled enterprises" (Yes/no/explain)	

Turkey

Name of survey	Community Innovation Survey 2018
Name of responsible agency	Turkish Statistical Institute, Sectoral Statistics Department
Reference period	2016-2018
Length of reference period	3 years
Reference year	
Data collection period	
First release of innovation results	
Concepts, notes and definitions in the questionnaire	
Differences with the OSLO manual	
Difference with the CIS questionnaire	
Target population cut-off point	10 or more employees
Industry coverage	Classification: ISIC Rev. 4/NACE Rev. 2. Core coverage: enterprises in the ISIC Rev. 4/NACE Rev.2 sections/divisions B, C, D, E, G46, H, J, K, M71-72-73.
Sampling frame used	
Completion requirement	
Unit of analysis surveyed	Enterprise
Survey type	
Type of interviews	
Average time to complete the survey	
Target population	
Target population (manufacturing)	
Target population (services)	
Achieved sample	
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	
Definition of strata	
Weights calculation methods: Variables used and methodology	
Implementation of non-response survey	
Number and types of reminders	
Use of imputation methods	
Combined with other surveys	
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	

United Kingdom

Name of survey	UK Innovation Survey
Name of responsible agency	Commissioning: Department for Business, Energy and Industrial Strategy. Survey design, field work and processing and estimation: Office for National Statistics (ONS) with boosts sample paid for by the Scottish Government.
Reference period	2016-2018
Length of reference period	3 years
Reference year	2019
Data collection period	January to October 2019
First release of innovation results	UK Innovation Survey 2019: Headline Findings https://www.gov.uk/government/statistics/uk-innovation-survey-2019-headline-findings 26 th March 2020
Concepts, notes and definitions in the questionnaire	Included within or immediately following the questions
Differences with the OSLO manual	None
Difference with the CIS questionnaire	All mandatory questions in CIS common questionnaire are applied to national questionnaire.
Target population cut-off point	10 or more employees
Industry coverage	See UK Innovation Survey 2019: Statistical Annex for details https://www.gov.uk/government/statistics/uk-innovation-survey-2019-main-report
Sampling frame used	National statistical business register
Completion requirement	Voluntary
Unit of analysis surveyed	Businesses
Survey type	Combined (The sample is a stratified design drawn from the Inter-Departmental Business Register (IDBR) with Neyman allocation used to determine the sample size in each stratum. Overall, roughly 15 per cent of the target population in sampled. A census for all large firms (250+ employees) is taken and a census of SMEs in SIC 72, one of the key policy areas is taken.
Type of interviews	Electronic questionnaire
Average time to complete the survey	60 mins
Target population	210,158
Target population (manufacturing)	
Target population (services)	
Achieved sample	14,040 out of 30,942 sampled
Achieved sample (manufacturing)	
Achieved sample (services)	
Weighted unit response rate	
Unweighted unit response rate	45.4%
Definition of strata	The sample is a stratified design drawn from the Inter-Departmental Business Register (IDBR) with Neyman allocation used to determine the sample size in each stratum. Overall, roughly 15 per cent of the target population in sampled. Stratification was based on three variables: 1) Region - All regions (9 Standard Regions in England) and countries in the UK 2) Division - Coverage of 25 specified sectors in the target population 3) Business Size: 10-49, 50-99, 100-249, 250 plus employees
Weights calculation methods: Variables used and methodology	The results are based on weighted data in order to be representative of the population of firms. The responses were

	weighted back to the total business population of those in the IDBR. On average each respondent represents 16 enterprises in the population.
Implementation of non-response survey	No
Number and types of reminders	2, postal
Use of imputation methods	
Combined with other surveys	No
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises" (Yes/no/explain)	No

United States

Name of survey	2019 Annual Business Survey
Name of responsible agency	US Census Bureau and US National Center for Science and Engineering Statistics / National Science Foundation
Reference period	2016-2018
Length of reference period	3 years
Reference year	2018
Data collection period	July 2019 – January 2020
First release of innovation results	
Concepts, notes and definitions in the questionnaire	Included in questionnaire
Differences with the OSLO manual	None. Although not all “core” tabulation questions are covered
Difference with the CIS quest.	
Target population cut-off point	Must have at least one domestic employee
Industry coverage	All domestic, non-farm, for-profit businesses.
Sampling frame used	National statistical business register
Completion requirement	Compulsory, but not enforceable in practice
Unit of analysis surveyed	Enterprise group
Survey type	Sample survey
Type of interviews	Electronic questionnaire
Average time to complete the survey	There is no separate estimate for response burden to complete just the innovation module, which is just one of several modules on the ABS.
Target population	5,260,000
Target population (manufacturing)	237,000
Target population (services)	5,020,000
Achieved sample	297,000
Achieved sample (manufacturing)	52,500
Achieved sample (services)	244,000
Weighted unit response rate	
Unweighted unit response rate	71.8 percent
Definition of strata	Strata by number of employees and industry.
Weights calculation methods: Variables used and methodology	Strata are defined by NAICS, geography, and demographic frame. The sampling weights are determined according to the sample size desired in each sampling stratum.
Implementation of non-response survey	There is very limited item non-response. In the 2019 ABS, there was no item non-response adjustment (so a missing answer to an innovation question is treated like a "No). A weight adjustment is also used for unit non-response.
Number and types of reminders	Three follow up mailings via the US Postal Service and email reminders when applicable
Use of imputation methods	There is very limited item non-response. In the initial (2018) ABS survey donor imputation was used for partial non-respondents to the R&D, innovation, and technology questions. In subsequent survey cycles there was imputation or no item non-response adjustment
Combined with other surveys	Yes
Link to documentation (outputs and questionnaire)	
Survey scope includes public (i.e. central, regional or local government-controlled) enterprises” (Yes/no/explain)	Yes